Yashaj Gupta

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Education

- **Jabalpur Engineering College** — Jabalpur, India
  B.Tech in Electronics and Telecommunication Engineering; GPA: 6.5 (Till Sem 6) — *2023 – 2027 (Expected)*

Skills

- **Programming**: C, C++, Embedded C, Python, SQL
- **Frameworks & Libraries**: Qt (Widgets/QML), OpenCV, TensorFlow
- **Embedded Platforms**: Linux, Arduino, STM32, ESP32, Raspberry Pi
- **Tools**: Git, CMake, MySQL, SQLite, Qt-SQL, Cadence OrCAD/Allegro PCB Designer, KiCad
- **Concepts**: Embedded Systems, PCB Design, IoT, RTOS Basics, UART, I2C, SPI, Data Structures & Algorithms
- **Soft Skills**: Leadership, Technical Writing, Public Speaking, Time Management

Experience

- **Noctilucent Technology** — On-site
  Embedded Systems Intern — *Dec 2025 – Feb 2026*
  - **Industrial System Development**: Developed a hazard detection and training board deployed at the **Reckitt** factory (Rudrapur).
  - **PCB Design**: Designed schematics and PCB layouts using **Cadence OrCAD/Allegro PCB Designer**, including component placement, routing, and design rule verification.
  - **Sensor Integration**: Integrated sensors with ESP32-based embedded hardware for real-time hazard detection and monitoring.
  - **Firmware Development**: Developed Embedded C/C++ firmware for sensor acquisition, monitoring, and alert generation.
  - **Testing & Debugging**: Performed hardware validation, PCB bring-up, firmware debugging, and system testing to ensure reliable industrial operation.

Projects

- **Smart Attendance System**
  ESP32, Raspberry Pi, Qt, SQLite, Wi-Fi — *Ongoing*
  - **Distributed Architecture**: Designed a building-wide attendance system using multiple ESP32 nodes connected to a Raspberry Pi central server.
  - **Desktop Software**: Developed a Qt desktop application for attendance monitoring, student management, and teacher authentication.
  - **Database**: Implemented centralized storage using SQLite/MySQL with real-time synchronization.
  - **Security**: Added role-based access control and secure communication between embedded devices and server.
- **Industrial Hazard Detection System**
  ESP32, Embedded C, OrCAD/Allegro PCB
  - **PCB Design**: Designed schematics and multilayer PCB using Cadence OrCAD/Allegro.
  - **Embedded Firmware**: Developed firmware for sensor acquisition and real-time alert generation.
  - **Deployment**: Built an industrial training board deployed at Reckitt Factory, Rudrapur.
- **Smart Canteen QR Payment System**
  ESP32, C++, QR Scanner, LCD, SQLite/MySQL, Wi-Fi
  - **Embedded System**: Developed a cashless payment system using ESP32 with QR code and PIN-based authentication for secure student transactions.
  - **Hardware Integration**: Interfaced QR scanner, LCD display, keypad, and RFID/Student ID modules for real-time user interaction.
  - **Security**: Designed secure authentication and real-time communication between embedded devices and the backend server.

Research Work

- **ABS (Anti-lock Braking System) Simulation**: **Stack:** MATLAB Simulink — Proposed an adaptive ABS model with slip-ratio control and dynamic hydraulic pressure distribution to enhance braking stability and safety.

Certificates

- **Introduction to C++ : Coding Ninjas (2024)**
- **Qt 6 Core with C++ : Udemy (2025)**

Volunteer Experience

- **Training and Placement Office, JEC** — India
  Assisted in managing and planning campus placement events — *Dec 2023 – Present*
- **IoT and Robotics Lab, JEC** — India
  Assisted in conducting IoT and robotics experiments and student project demonstrations — *July 2025 – Present*